Artificial intelligence

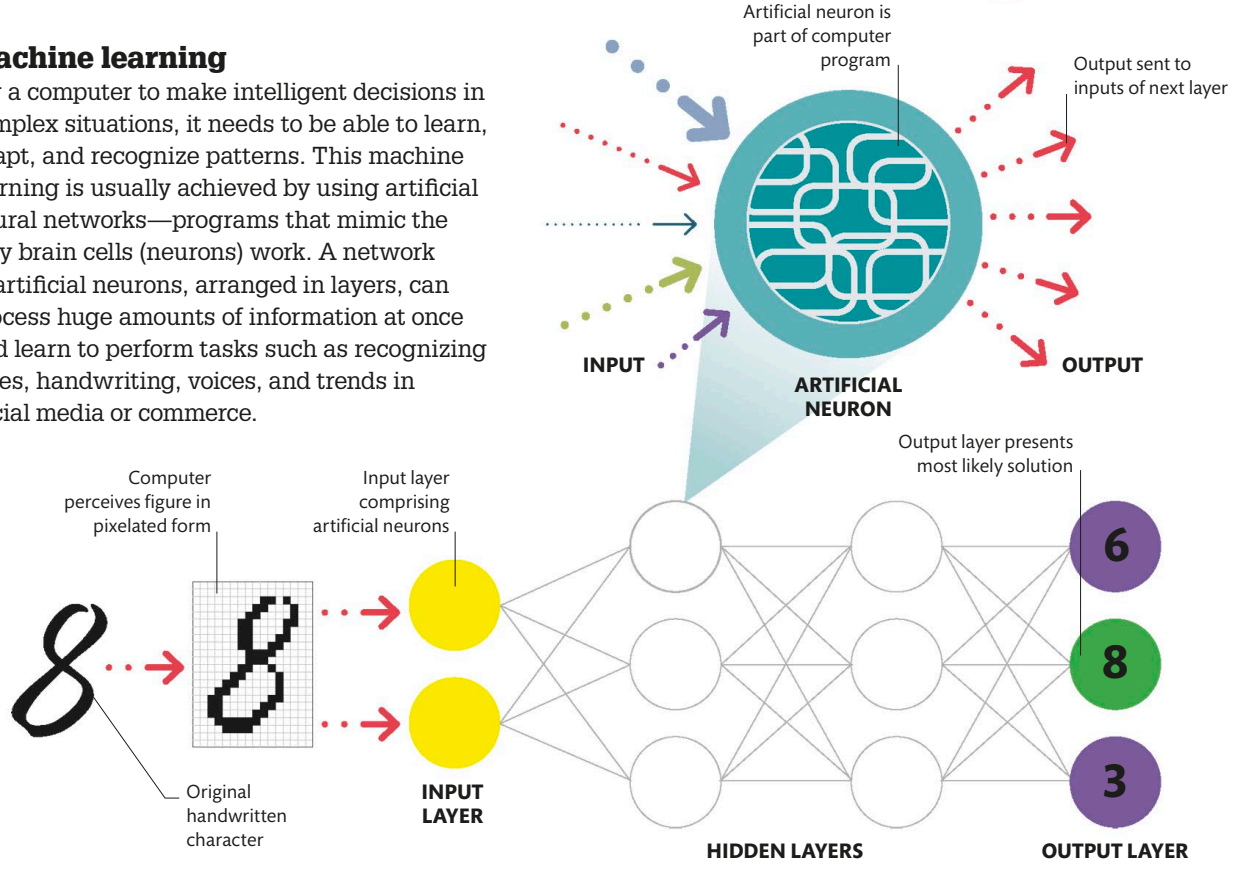
Artificial intelligence (AI) is any technology that enables computers to do things that humans consider intelligent, such as recognizing patterns and solving problems. One goal of AI is for computers to "think" for themselves—to make their own decisions and respond to situations.

HOW DOES SPEECH RECOGNITION WORK?

A computer can recognize the building blocks of speech, called phonemes, and work out the words it has heard spoken.

Machine learning

For a computer to make intelligent decisions in complex situations, it needs to be able to learn, adapt, and recognize patterns. This machine learning is usually achieved by using artificial neural networks—programs that mimic the way brain cells (neurons) work. A network of artificial neurons, arranged in layers, can process huge amounts of information at once and learn to perform tasks such as recognizing faces, handwriting, voices, and trends in social media or commerce.



Artificial neural network
Real neurons produce outputs based on inputs they receive from the senses and from other neurons—but they can change how they respond over time, depending on the inputs. Artificial neural networks work in the same way, and, like real ones, they are arranged in layers.

Input layer
The initial layer receives inputs. In this example, each neuron receives a number representing the brightness of an individual pixel from a digitized image of a handwritten character. Only two input neurons are shown here, but a real system would have many more.

Hidden layers
The output of each neuron in the input layer is also a number, the value of which depends on that of the input multiplied by a "weight." The weight changes as the network learns. The number passes on to neurons in several layers, each with its own weight.

Output layer
The outputs of the neurons in the hidden layers are passed to neurons in the output layer. In this network, there would be 10 output neurons—one for each of the digits 0 to 9. The network's "guess" for the character is the neuron that has the highest weighting.